ABHISHEK KUMAR

AI Native Developer & AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 2.5+ years of experience deploying production AI systems across NLP, Vision, Speech, and Data Engineering. Spearheaded real-time speaker identification, RAG pipelines, STT/TTS microservices, and compliance tools. Reduced manual workflows by 65% and architected REST microservices. Expertise in LangChain, FastAPI, PyTorch, Whisper, YOLO, Docker, Qdrant, and AWS.

PROFESSIONAL EXPERIENCE

Softel Solutions Pvt. Ltd | *Python Developer (AI/ML)* **Mar 2026 - Present**
Noida, India

- **Real-time Speaker Identification Module:** Spearheaded deploying a voice biometric system using **PyTorch, ECAPA-TDNN (Wav2Vec2),** and **ChromaDB vector database**, achieving 94% accuracy across 250+ users.
- **Enterprise Hybrid RAG & GraphRAG Platform:** Scaled a hybrid retrieval platform serving 15K+ daily queries for 1,200+ users via FastAPI, Neo4j, ChromaDB, and Redis RQ. Evaluated on a 400-query test set (Ragas), cutting hallucinations by 22% and optimized latency by 35%.
- **Geospatial & Atmospheric Data Pipeline:** Engineered an ingestion engine (asyncio/multi-threading), processed 100+ atmospheric parameters and satellite tiles into MongoDB with 10,000-doc bulk writes, improved throughput by 40%.
- **Speech-to-Text (STT) & Text-to-Speech (TTS):** Developed audio microservices using **Whisper** and **Bark/Coqui TTS**, containerizing services via FastAPI and Docker for production deployment.

Azine Web Technologies Pvt. Ltd | *Machine Learning Engineer* **Feb 2025 - Feb 2026**
Ahmedabad, India

- **AML Adverse Media & Entity Risk Platform:** Architected an automated entity risk assessment system featuring multi-source adverse media screening (APC), PEP/SAN/KYB checks, and a weighted 0-100 risk scoring engine across 30,000+ entities.
- **Compliance & News Intelligence Automation:** Spearheaded integration of 200+ OpenSanctions datasets, live news feeds, DistilBERT, and SVM sentiment classifiers, increased high-risk detection by 28% and reduced effort by 65%.
- **Real-time Vision & Identity Verification:** Implemented and deployed real-time identity verification APIs (liveness detection, signature recognition) and barcode extraction pipelines using **YOLOv11, OpenCV,** and **FastAPI**, achieving 95%+ field extraction accuracy.
- **Automated Data & Scraping Pipelines:** Delivered ETL web scraping pipelines extracting compliance data across 150+ government portals using Selenium, BeautifulSoup, and PDFPlumber on 3-day cron cycles.

Phoenix Labs Global | *AI & ML Intern* **Jul 2024 - Jan 2025**
Remote

- **Talk2DB - Natural Language to SQL/Mongo Agent:** Pioneered an AI database query engine using **FastAPI, Mem0 (schema memory),** and **OpenAI**, converting natural language to validated SQL/Mongo queries across PostgreSQL, MySQL, and MongoDB.
- **Road Infrastructure Vision & Inspection System:** Engineered a vision inspection pipeline processing dual-camera video feeds (PAVE & ROW cameras) with automated pavement distress classification and Rut Depth / IRI rating evaluation.
- **50-Camera Real-Time Facial Recognition System:** Designed a biometric surveillance engine processing 50 concurrent CCTV video feeds (RTSP/non-RTSP), using **RetinaFace, ArcFace embeddings,** and **Qdrant vector search** (<5ms query latency across 1M+ faces).
- **API Orchestration Gateway:** Orchestrated a Django API gateway managing 13+ FastAPI ML microservices with RBAC, and anomaly detection engines, reduced false positives by 37%.

Atharvo India Pvt. Ltd | *Machine Learning Intern* **Jan 2024 - Jun 2024**
Remote

- **Handwritten Digit Recognition (MNIST):** Assembled a LeNet-5 architecture using **TensorFlow** to classify handwritten digits, focusing on training efficiency and model generalization. Programmed an end-to-end ML pipeline, including preprocessing, training, validation, and comparative accuracy benchmarking.

- **Churn Prediction & Anomaly Modeling:** Trained ML models (XGBoost, Random Forest, Scikit-learn) to predict customer churn, achieving 91% accuracy and saving 15% retention costs.
- **NLP Ticket Classification:** Structured a text classification pipeline using TF-IDF and Logistic Regression, improving ticket routing by 30%.

TECHNICAL SKILLS

Languages & DBs	Python (Advanced) • SQL • Java • JavaScript • MongoDB • PostgreSQL • MySQL • Redis • Qdrant • ChromaDB • Neo4j GraphRAG
Audio & Speech AI	Wav2Vec2 • ECAPA-TDNN • Whisper (STT) • Bark (TTS) • Coqui TTS • Speaker Recognition • Audio Diarization
Generative AI & LLMs	LangChain • LangGraph • LlamaIndex • Mem0 (Vector Schema Memory) • RAG • Vector Embeddings • Transformer Architecture
ML & Computer Vision	PyTorch • TensorFlow • Scikit-learn • DistilBERT • SVM • YOLOv11 • RetinaFace • ArcFace • OpenCV • PaddleOCR • Tesseract
DevOps & Engineering	FastAPI • Django • Flask • REST API • Microservices • Docker • AWS (EC2, S3) • Git • Selenium • BeautifulSoup • Celery • RQ Workers

EDUCATION

Bachelor of Technology (B.Tech) - Electronics & Communication Engineering — **Sep 2020 - Jun 2024**
Oriental Institute of Science and Technology, Bhopal, India — ***CGPA: 8.17 / 10***

CERTIFICATIONS

- **AWS & Cloud:** Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (Mar 2025)
- **NLP & Deep Learning:** NLP Mastery in Python - Udemy (Jul 2025), Deep Learning for Developers - Infosys (Aug 2024)
- **Machine Learning:** ML Internship - Infosys Springboard (Dec 2024), Data Science & ML BootCamp - Udemy (Jul 2025)
- **Core Development:** Java & Python Full Stack - (Dec 2024), SQL for Data Science - UC Davis/Coursera (May 2023)